Time Series Practice Questions

1. You are given the following time-series for monthly building settlement (mm):
   5, 7, 10, 12, 15, 14, 12, 9, 7, 6, 7, 9
   Identify whether each of these components exist in this time series: trend, seasonality and noise. Approximately sketch the decomposition.

2. You are working on modelling displacement of bridge girder. You have collected the following data:

$$t: 0,1,2,3,4$$
$$x(t): 10,13,15,18,20$$

   Assuming linear trend model $x(t) = a + bt + \epsilon_t$, find the Least-Square Estimate for a and b. What type of error are we implicitly assuming here?

3. Given the following observations:

$$y = Ax + \epsilon$$
$$y = \begin{bmatrix}5\\8\\6\end{bmatrix}, A = \begin{bmatrix}1\\1\\1\end{bmatrix}, cov(\epsilon) = \begin{bmatrix}4 & 1 & 0\\1 & 4 & 1\\0 & 1 & 4\end{bmatrix}$$

   Compute the BLUE of x.

4. In question 3, compute the LS estimate. How do the two compare?